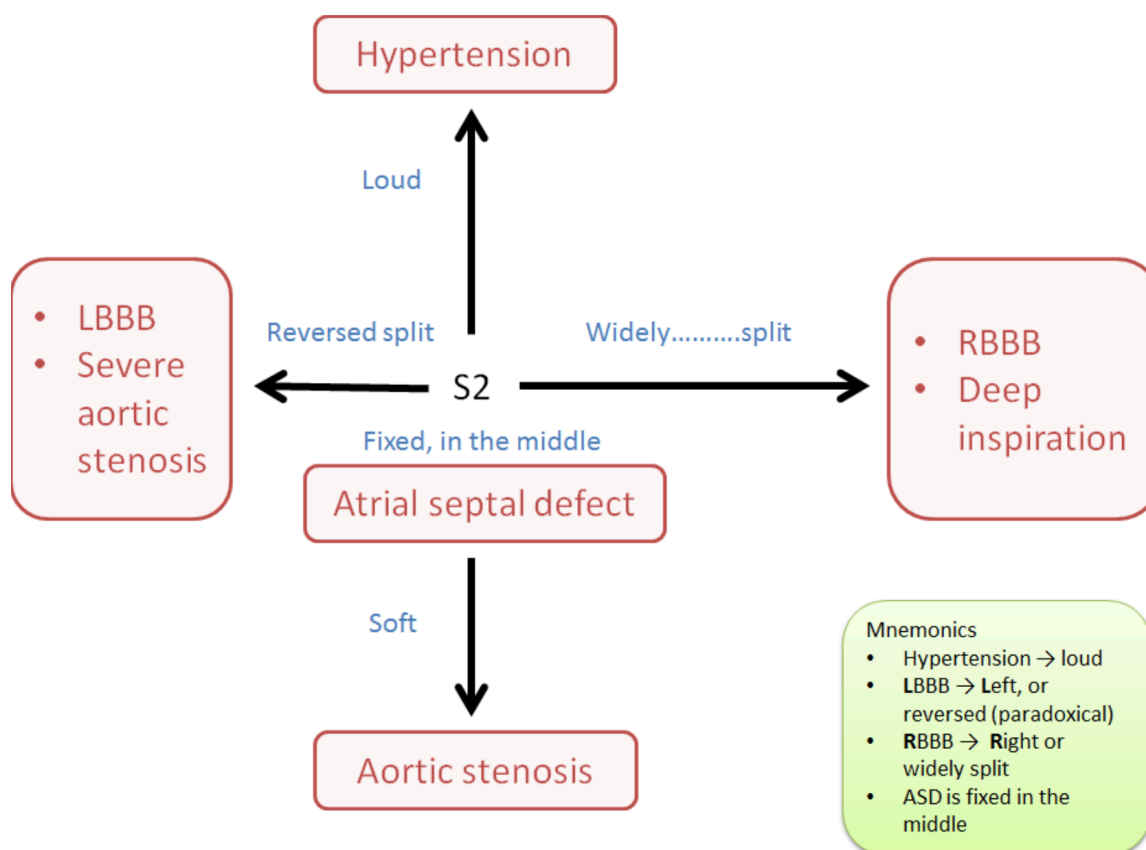


Valvular Heart Diseases:

Aortic Stenosis (calcification of aortic valve):

- Ejection systolic murmur radiating to the carotid
- Quieter with left ventricular systolic dysfunction. (Also with valsalva)
- Fx: narrow pulse pressure, **slow rising [4x wrong]**, delayed ESM, **soft S2**
- **Severe stenosis: soft second heart sound / displaced apex beat / S4 (heart sound before S1) [S4 3x wrong]**
 - **S4 coincides with the P waves on an ECG [2X WRONG]**
- MC cause:
 - Young (<65): bicuspid aortic valve
 - Older (>65): calcification
- Management:
 - **Replacement if:**
 - **symptomatic (exertional dyspnea)**
 - **peak gradient >40**
 - **valve area of <1**



Aortic Regurgitation:

- Associated with **rheumatic fever, syphilis, Marfan's syndrome, ankylosing spondylitis**, bicuspid aortic valve and post-rheumatic disease.
- **Murmur: early diastolic murmur [4x wrong]**
- Abrupt distention and collapse of the carotid arteries along
- visualisations of capillary pulsations with light compression applied to fingernail bed.

Pulmonary Stenosis:

- Ejection systolic, louder on inspiration.

Mitral Stenosis:

- **Mid diastolic rumbling murmur** best heard at the apex
- Sx: hemoptysis, dyspnea
- Severity indicated by a longer murmur
- Tapping apex beat and parasternal heave association.
- Management:
 - **Percutaneous Mitral commissurotomy if severe** (secondary to rheumatic) [2x wrong]

MR: mitral clip insertion

- Pan-systolic murmur louder during expiration
 - Heavily calcified valve, significant MR or anatomical consideration: MV replacement

MVP is associated with:

- Osteogenesis imperfecta, pseudoxanthoma, turner's and marfans.

Tricuspid regurgitation:

- **Systolic murmur louder during inspiration. [2x wrong]**
 - **Left parasternal heave is a feature [3x wrong]**

PDA is associated with a **collapsing pulse** (left to right shunt)

- Ibuprofen or indomethacin is given if still after 1 week of birth
- Apex beat due to left ventricular overload.
- Wide pulse pressure
- Left subclavicular thrill

Diagnosis & notes	RA	RV	PA	LA	LV	Ao
Normal	70%	70%	70%	100%	100%	100%
Atrial septal defect (ASD) The oxygenated blood in the LA mixes with the deoxygenated blood in the RA, resulting in intermediate levels of oxygenation from the RA onwards	85%	85%	85%	100%	100%	100%
Ventricular septal defect (VSD) The oxygenated blood in the LV mixes with the deoxygenated blood in the RV, resulting in intermediate levels of oxygenation from the RV onwards. The RA blood remains deoxygenated	70%	85%	85%	100%	100%	100%
Patent ductus arteriosus (PDA) Remember, a PDA connects the higher pressure aorta with the lower pressure PA. This results in only the PA having intermediate oxygenation levels	70%	70%	85%	100%	100%	100%
VSD with Eisenmenger's	70%	70%	70%	100%	85%	85%
PDA with Eisenmenger's	70%	70%	70%	100%	100%	85%
ASD with Eisenmenger's	70%	70%	70%	85%	85%	85%

VSD is associated with right heart failure, aortic regurgitation, eisenmenger's complex, and infective endocarditis.

Oxygen saturation for cardiac catheterization:

Etc:

- Antibiotic prophylaxis to prevent infective endocarditis is NOT recommended for dental procedures instead general oral hygiene advice is given.

Rheumatic heart fever:

- Histology: Aschoff bodies & Anitschkow cells

Valve replacements:

- Young / cardiomyopathy / endocarditis / rheumatic fever: mechanical valve
- Childbearing woman / bleeding RF: prosthetic valve

Congenital heart diseases:

- Cyanotic —> **TGA MC at birth**, but overall **Fallot's most common**
- Acyanotic —> VSD most common

Tetralogy of Fallot:

- Associated with the following:
 - Right ventricular outflow tract obstruction
 - Right to left shunt
 - Right ventricular hypertrophy
 - Ejection systolic murmur
 - Over-riding aorta

Transposition of great vessels:

- Due to the failure of the aorticopulmonary septum to spiral

Blue Toe Syndrome:

- Cholesterol embolisation

Acute Pericarditis:

Etiology:

- Viral, TB, uremia
- Post MI
- Radiotherapy
- CT diseases (SLE, RA)
- Hypothyroidism
- Lung and breast CA
- Trauma

ECG Changes: global widespread saddle-shaped ST elevations, **PR depressions**.

Management:

- NSAIDs and Colchicine first-line for 1-2weeks
- Inpatient only if high fever or elevated troponin
- Avoid strenuous activity
- Hemodialysis if uremic pericarditis.

Hypertension:

Any reading above 140/90: offer ABPM

A. FIB:

CHA2DS2-VASc:

C: congestive heart failure

H: hypertension

A2: age 75 or above (2), or age 65-74

D: DM

S2: prior stroke, TIA, or thromboembolism

V: vascular disease

S: sex female

Anticoagulation:

0: no treatment

1: males consider anticoagulation, female no treatment

2: offer anticoagulation (DOACs) — apixiban, dabigatran

Rate Control:

- Beta blockers
- CCBs in the case of contraindication to asthma
- Digoxin

Lifelong warfarin —> 2 or more eps

4 weeks warfarin and review —> 1 ep

Idarucizumab is the reversal agent for dabigatran.

Cardiomyopathy:

Dilated:

- Caused MC idiopathic or by coxsackie, HTN, hemochromatosis, or alcohol.
- Can be inherited (DMD)
- Can be nutritional (Thiamine deficiency), or even Selenium deficiency (Keshan disease)
- Balloon appearance of the heart on CXR
- Managed with HFrEF management (ACE, BB, MRA, SGLT2)

Restrictive:

- MC cause: Amyloidosis,

HOCM: AD

- **CCC: systolic anterior motion of the mitral valve**
 - Mutation in the gene encoding Beta myosin heavy chain protein OR myosin binding protein C.
- Examination: double apex beat and a harsh systolic murmur
- **Poor prognosis → septal wall thickness of >3cm (risk of SCD)**
- Marker for sudden death: abnormal blood pressure on exercise
- Biopsy: myofibrillar hypertrophy, myocyte disarray, and interstitial fibrosis.
- ECG: deep q waves in lateral leads, non-specific t wave inversions.
- Echo: systolic anterior motion of the anterior mitral valve leaflet, with **mitral regurgitation.**
 - **Left ventricular hypertrophy + outflow tract obstruction**
- **AVOID giving ACE inhibitors (Worsens outflow tract obstruction)**
- **Management: Implantable cardioverter defibrillator**
- **MC cause of death: ventricular arrhythmia.**

Arrhythmogenic right ventricular cardiomyopathy (ARVD):

- **CCC: right ventricular myocardium replaced by fatty and fibro-fatty tissue.**
- AD inherited
- ECG: most specific CCC findings is small positive deflection at the end of the QRS complex.
 - T wave inversion in V1 - V3
- AR variant: Naxos disease (ARVC + woolly hair + palmoplantar keratosis)

Infective Endocarditis:

Poor prognostic factors:

- **S. Aureus infection**
- **Prosthetic valve**
- **Negative culture**
- **Low complement levels**

Good prognostic factor:

- **S. Viridans**

S. Aureus is the MC cause (esp in iv drug users)

S. Epidermidis: MC cause if <2mths post valve surgery

Poor dental hygiene —> Viridians (e.g. Sanguinis / Mitis)

Management:

- Blind therapy:

- **Native —> Amoxicillin (If allergy: Vanco) + - gentamicin [2x wrong]**
- **Prosthetic —> Flucloxacillin (or Vanco) + Rifampicin + low dose gentamicin [3x wrong]**
- Native valve due to staphylococci: Flucloxacillin (If allergy: Vanco + Rifampicin)
- Prosthetic valve due to staphylococci: Flucloxacillin + Rifampicin + low-dose gentamicin
- **Due to Viridans: Benzylpenicillin**

Surgical indications:

- Severe valvular incompetence
- Aortic abscess (indicated by lengthening of PR interval)
- Infections resistant to antibiotics
- Cardiac failure refractory to medical treatment (congestive symptoms)
- Recurrent emboli after antibiotics.

Modified Duke Criteria:

- Dx if —> 2 major // 1 major + 3 minor // 5 minor // pathological criteria
- Pathological: +ve histology
- Major: 2 +ve blood cultures, endocardial involvement (+ve echo — abscess, valvular regurgitation)
- Minor: predisposing heart condition, IV drugs, fever >38, major emboli, splenomegaly, clubbing, splinter hemorrhage, janeway lesions, purpura, glomerulonephritis, Osler's nodes, Roth spots

Angina:

Medications:

- **Aspirin + Statin**
- **Sublingual GTN**
- **Beta blocker** or CCB (BB added first)
 - If **CCB** alone: Verapamil or Diltiazem
 - If combo: Nifedipine or Amlodipine (Also if poor response to max dose beta blocker)
 - If STILL: add a third drug then PCI

Hyperlipidaemia:

Drugs	MOA	Adverse effects
Statins	HMG CoA reductase inhibitors	Myositis and deranged LFTs
Fibrates	Agonist of PPAR-alpha increasing lipoprotein lipase expression	Myositis, pruritic and cholestasis
Nicotinic acid	Decreases hepatic VLDL secretion	Myositis, flushing
Ezetimibe	Decreases cholesterol absorption in the small intestine	Headaches
Cholestyramine	Decreases bile acid reabsorption in the small intestine (up regulating amount of cholesterol converted to bile acid)	GI side effects

Myocardial Infarction:

Arterial supply of the heart:

- Left aortic sinus —> LCA
- Right aortic sinus —> RCA
- LCA —> LAD + circumflex
- RCA —> posterior descending
- RCA supplies SA node in 60%, AV node in 90%

Complications:

- Cardiac arrest (**MC**), **due to V. Fib**
- Cardio-genic shock
- Chronic heart failure (ventricular ejection fraction is the most important predictor)
- Tachy-arrhythmias (V. Fib)
- Brady-arrhythmias (AV Block)
- Pericarditis (in the first 48hrs)
- Dressler's syndrome)2-6 weeks)
- Left ventricular aneurysm
- Left ventricular free wall rupture
- VSD
- Acute MR (early systolic murmur)
- Rupture of papillary muscle (associated with acute mitral regurgitation [widespread systolic murmur + hypotension + pulmonary edema])

Right ventricular systolic dysfunction + right ventricular free wall dyskinesia —> right ventricular myocardial infarct (NITRATES AVOIDED DUE TO REDUCING PRELOAD)

Complete heart block following MI (assessed by pallor, bradycardia, hypotension) —> right coronary artery affected.

DVLA & CVS disorders:

- Catheter ablation: 2 days off driving
- **Angioplasty (elective): 1 week off driving**
- **Pacemaker insertion: 1 week off driving**
- CABG: 4 weeks off driving
- **ACS: 4 weeks off driving**
- Heart transplant: 6 weeks do not drive, no need to notify.

- ICD:
 - If for sustained ventricular arrhythmia: cease driving for 6 months
 - Loss of HGV license
 - **If prophylactic:** for 1 month

- Hypertension: If resting BP >180
- Angina: if symptoms at rest or whilst driving
- Aortic aneurysm of 6cm or more: annual review
 - >6.5cm: permanent ban

Medication Information:

For pulmonary arterial hypertension (pulmonary arterial pressure more than 25) with **positive response to vasodilator testing [4x wrong]: CCB prescribed.**

- **if negative: Sildenafil (phosphodiesterase inhibitor) [2x wrong]**

- **If positive: CCBs (Nifedipine)**

* **Decrease in pulmonary arterial pressure upon N.O. inhalation: positive response**

Heparin induced thrombocytopenia: anticoagulation given (Argatroban infusion)

- It is a prothrombotic condition that occurs 5 - 10 days of starting treatment.
- Diagnosis requires serotonin release assay or **antibodies against Heparin PF4 antibody (wrong 3x) ELISA test**
- Treatment is discontinuation of all forms of heparin and starting non-heparin anticoagulation.

Sotalol (class III anti-arrhythmic drug) **blocks K channels and direct effect on QT interval.**

Adenosine is contraindicated in asthma, instead verapamil is given.

Phenelzine has a high risk of hypertensive crisis.

Diltiazem (and other CCBs are avoided in patients with HF and reduced EF) —> negative inotropic effects.

Pravastatin causes more myopathy than Simvastatin.

Amlodipine: gingival hyperplasia

Aspirin can be safely taken in dental practice.

Warfarinised patient: check INR 72 hrs before dental procedure, and proceed if INR <4.

Statins are contraindicated in pregnancy.

- They also interact with grapefruit juice

Verapamil is NOT used in V. Tach and precipitates V. Fib.

- Side effect: constipation (all CCBs)

Ivabradine is associated with visual disturbances (phosphenes and green luminescence [bright vision spots])

The use of beta blockers has decreased over the past years due to less likely to prevent stroke and potential impairment of glucose tolerance.

Warfarin:

- Can be used in breastfeeding
- Target INR for any VTE remains 2 - 3 (2.5)

Benzylpenicillin & gentamicin for endocarditis: monitor PR interval

Thiazides act on the proximal part of the distal convoluted tubule (inhibited Na reabsorption by blocking Na/Cl symporter)

- Hypokalemia is due to **inhibition of sodium reabsorption by blocking the symporter at the beginning of the distal convoluted tubule...** thus sodium that reaches the collecting ducts is increased. [another cause includes activation of the renin-angiotensin-aldosterone system secondary to hypovolemia]
- Side effects: pancreatitis, impotence, agranulocytosis, photosensitivity rash.
 - hypokalemia, hyponatremia, hypercalcemia,
 - They also may cause gout
 - Worsen glucose intolerance

Erythromycin prolongs QT

Atropine —> Mydriasis

Amiodarone —> hypokalemia

- Half life: 20 to 100 days

ISMN is given at asymmetrical dosing to prevent nitrate tolerance.

Moxonidine is a central acting antihypertensive drug.

Hydralazine is a direct acting vasodilator

Sodium nitroprusside is a venous and arterial smooth muscle.

Diazoxide is a vasodilator

Digoxin toxicity: hypernatremia, hypokalemia, hypomagnesemia, hypothermia, HYPERcalcemia.

- In elderly: reduced eGFR

- Shortens QT interval

Ticagrelor treated patients can have dyspnea after an MI (due to impaired clearance of adenosine)

Nicorandil is a substitute for beta-blockers in the case of angina (e.g. when patient is also on

verapamil — we do not give bb with verapamil due to bradycardia)

- causes ulceration in the GI Tract (anywhere in the GI tract)

Before starting Flecainide look for structural heart diseases in an echo.

Clopidogrel is less effective from Omeprazole.

ACE inhibitors can cause angioedema.

Lidocaine is a Class Ib agent

Adrenaline doses: [review]

- Anaphylaxis: 0.5ml 1:1,000 im

- Cardiac arrest: 10mL 10,000 IV or 1mL of 1:1000 IV // 1mg 1:10,000

- —> antidote for epinephrine: phentolamine

- —> increase lactate production if infusion

Drug	MOA
Heparin / Fondaparinux	Activates anti-thrombin III [2x wrong]
Clopidogrel & Ticagrelor	P2Y ₁₂ inhibitor (reversible ADP binding to its platelet receptor) [3x wrong]
Abciximab	Glycoprotein IIb / IIIa inhibitor
Dabigatran / Bivalirudin	Direct thrombin inhibitor
Rivaroxaban	Direct factor X inhibitor
Aspirin	Irreversible inhibition of COX 1 and 2 (preventing conversion of arachidonic acid to prostaglandins to prevent thromboxane A₂ formation in platelets [3x wrong]—> which promotes platelet aggregation).
Statins	Inhibit action of HMG-CoA reductase (rate-limiting enzyme in hepatic cholesterol synthesis)
ARBs	Prevents angiotensin 2 acting on AT ₁ receptors

Drug	MOA
ACE	Dilation of EFFERENT glomerular arterioles / Bradykinin accumulation
Class I anti-arrhythmic drugs	Blocks Nav1.5 Na channels to slow upstroke of cardiac action
Endothelial receptor antagonists (___sentan)	Decrease pulmonary vascular resistance - Used in pulmonary arterial hypertension [2x]
Antiplatelet agents (Pyridamole)	Phosphodiesterase inhibitor, decrease adenosine uptake.
Thiazides	Inhibition of Na/Cl supporter at the distal convoluted tubule.
Clexane	Activation of anti-thrombin III which inhibits factor Xa
Sacubutril + ARB	Prevent degradation of BNP and ANP
Glycoprotein antagonists (Tirofiban)	Reversible antagonist of GP IIb / IIIa receptors
Dipyridamole	Phosphodiesterase inhibitor (INCREASES effect of adenosine)
Diuretics (Furosemide, _tanide)	Inhibits Na-K-Cl cotransporter in the ascending limb of loop of hence [2X WRONG]
Amiodarone	Blocks potassium channels - Long half-life (thus loading dose is given)
Flecainide	Sodium channel blocker (Nav 1.5 block)
Digoxin	Sodium-potassium ATPase pump inhibitor Given when needed to rate control
Amiloride	Epithelia sodium channel inhibitor in the distal convoluted tubule (potassium sparing diuretic)
Nicorandil	Potassium channel activator
Nitrates (isosorbide)	Release of NO in smooth muscle, activating guanylate cyclase which converts GTP to cGMP = smooth muscle relaxation
Alteplase	Catalyses the conversion of plasminogen to plasmin

Antiarrhythmics: Vaughan Williams classification ★

The Vaughan Williams classification of antiarrhythmics is still widely used although it should be noted that a number of common drugs are not included in the classification e.g. adenosine, atropine, digoxin and magnesium

AP = action potential

Class	Examples	Mechanism of action	Notes
Ia	Quinidine Procainamide Disopyramide	Block sodium channels	Quinidine toxicity causes cinchonism (headache, tinnitus, thrombocytopenia)
		Increases AP duration	Procainamide may cause drug-induced lupus
Ib	Lidocaine Mexiletine Tocainide	Block sodium channels Decreases AP duration	
Ic	Flecainide Encainide Propafenone	Block sodium channels No effect on AP duration	
II	Propranolol Atenolol Bisoprolol Metoprolol	Beta-adrenoceptor antagonists	
III	Amiodarone Sotalol Ibutilide Bretylium	Block potassium channels	
IV	Verapamil Diltiazem	Calcium channel blockers	

ECG Information:

For **Hypokalemia** ECG: peaked T waves, **long QT syndrome**

- **severe: U waves [2x wrong], ST depression, and TWI**
- Sertraline avoided in long QT syndrome (antidepressant exacerbate these arrhythmias)

Hypocalcemia: long QT waves

Hypercalcemia (nausea, abdominal pain, general muscle pain and weakness): shortened QT waves

* **QT is measured from start of Q to the end of the T**

- * **Long QT + deafness: Jerrell and Lange Nielsen syndrome (AR mutation in KCNQ1 or KCNE1)**

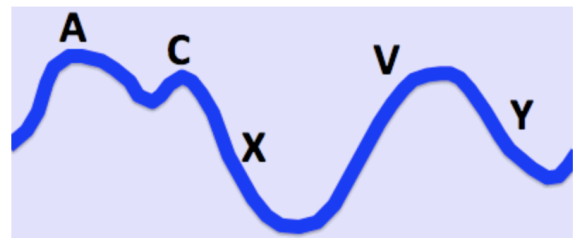
Left axis deviation	Right axis deviation
Ostium primum ASD	Ostium secundum ASD
Obesity	Left posterior hemiblock
Left BBB	Right ventricular hypertrophy
Left anterior hemiblock	Pulmonary embolism
WPW syndrome + R sided accessory pathway	Chronic lung disease

JVP needs extensive review, multiple times wrong.

In JVP: Exaggerated V waves (with tricuspid regurgitation).

A wave —> Atrial contraction

- **Cannon: single chamber ventricular pacing**
 - **Regular:** atrial contractions against closed tricuspid valve (AKA AV nodal re-entry tachycardia)
 - **Irregular: Complete heart block OR.. V. Tach**



- Absent A waves: Atrial fibrillation
- Large A waves: pulmonary stenosis (harsh systolic murmur) / HOCM / tricuspid stenosis

C wave —> Closure of tricuspid valve (not normally visible)

V wave —> Passive blood filling into atrium against a **closed tricuspid valve**

- **Giant (exaggerated/prominent): decompensated heart failure (right-sided) Tricuspid regurgitation**
- Absent V waves: severe hypovolemia or complete heart block

X wave —> fall in atrial pressure during ventricular systole (x4 wrong)

- Exaggerated X Descent: Constrictive pericarditis and cardiac tamponade
 - **NO Y DESCENT IS characteristic for tamponade**

Y wave —> opening of tricuspid valve

- Slow Y descent: tricuspid stenosis
- Exaggerated Y Descent: constrictive pericarditis (

AV dissociation (P waves dissociated from QRS and less):

—> **complete heart block: variable intensity of S1**

—> **ALSO V TACH**

J waves: Hypothermia

Q waves: Previous MI

Delta waves: WPW

Persistent ST elevation follow recent MI with NO chest pain: left ventricular aneurysm

Torsades De Pointes: polymorphic ventricular tachycardia with a prolonged QT interval.

- Associated with: TCAs/ SAH, Hypothermia
- Associated with Romano-Ward Syndrome (most common inherited form of long QT syndrome (LQTS))
- **Risk factor: Clarithromycin (known to prolong QT interval), also Risperidone.**
- Management: IV Magnesium Sulfate

Positive QRS concordance in chest leads is associated with v. Tach

Etc Information:

Primary cardiac tumors MC site: left atrium (Atrial Myxoma)

Myoglobin is the first that rises in the myocardial infarction.

For multifocal atrial tachycardia: Verapamil is given

Myxomas are associated with:

- Clubbing
- Mid-diastolic murmur
- Pyrexia
- Atrial Fibrillation

Restrict fluids in Pre-eclampsia (pulmonary and cerebral edema)

- If previous pregnancy: low dose aspirin given in next pregnancy

ACLS in pregnant women beyond 20 weeks: position with a left lateral tilt of 15 degrees.

Pulmonary embolism:

- Unprovoked: continue for 6 months
- Provoked: 3 months

Troponin controls cardiac muscle contraction and relaxation, it is a component of thin filaments.

- Binds to tropomyosin

Electrical cardioversion is synchronized to the R wave

For prosthetic heart valves:

- **If bio-prosthetic: must be on aspirin (x3 wrong)**
- If **mechanical: must be on warfarin and aspirin**
 - **Lower failure rate than bio-prosthetic**
 - For warfarin: lifelong
 - **Aortic valve: INR target of 3.0**
 - **Mitral valve: 3.5 [2x wrong]**

SVT not relieved by vagal or adenosine → verapamil given (likely AVNRT)

In pregnancy: HTN falls in the first half of pregnancy before rising to the pre-pregnancy levels before term.

Fluorodeoxyglucose if the radio tracer used during PET scans.

Gallium is used in nuclear medicine for valium scans (tumor activity)

Technetium 99mTc IS THE mc radionuclide IN NUCLEAR MEDICINE, used in PSECT studies.

Thallium is used in myocardial perfusion imaging.

Palpitations are investigated with 72hr Holter monitors.

Migraine + Stroke = Patent foramen ovale

- PFO is the MC cause of any embolus (e.g. DVT)

DVT confirmation in pregnant: compression duplex Doppler

Ebstein's anomaly: tricuspid regurgitation (**pan [holo] systolic murmur worse on inspiration**) **[3x wrong]**

- Presents with **arrhythmia and SOB on exertion**
- Associated with lithium in pregnancy.
- ECG: RBBB and WPW syndrome due to aberrant conduction pathways.
- Giant V waves on JVP

WPW is associated with mitral valve prolapse, Ebstein's anomaly, thyrotoxicosis and HOCM.

- **Wide QRS complexes with a slurred upstroke.**
- **Verapamil is contraindicated and dangerous in WPW.** (Blocks AV node without any effect on accessory pathway precipitating V. Fib)
- Medications: Sotalol, Amiodarone, **Flecainide**
- Definitive treatment: radio frequency ablation of accessory pathway

AKI following angiography is associated with cholesterol emboli (high Urea and high Eosinophils).
- Present with livedo reticularis (lace-like rash)

Ventricular arrhythmia following an angiogram is a transient complication and often normal. Safe to discharge afterwards.

Eruptive xanthomas are associated with familial hypertriglyceridaemia

MC cause of endocarditis <2months post-valve surgery: Staphylococcus epidermidis

Colorectal CA is associated with Streptococcus Bovis (Gallolyticus)

Poor dental hygiene is associated with endocarditis **secondary to Viridians (Sanguines)**

Pulmonary capillary wedge pressure is a measurement through Swan-Ganz catheter. It provides an indirect estimation of left atrium pressure.

During exercise:

- Venous constriction
- 3 fold increase in HR
- 50% increase in stroke volume
- Increased myocardial contractibility
- Stable / Slight decrease in DBP

Takayasu Arteritis: young Asian woman with headache, malaise, joint pain, claudication symptoms, and significant blood pressure between arms.

- Obliterative arteritis affecting **the aorta**
- Associated with AR (carotid bruit presence + diastolic murmur).

Catecholaminergic polymorphic ventricular tachycardia:

- Symptoms develop before the age of 20 years
- Resting ECG is normal (diagnosis is on bidirectional VT during exercise testing)
- Beta blockers are the first line treatment
- Most are due to defects in the calcium channels.

Brugada syndrome: mutation in the **myocardial sodium ion channel proteins.**

- Febrile, young, southeast Asian.... Or **presents with fainting feeling.**
- ECG finding: convex **ST elevation in V1 to V3** (Brugada type 1 pattern)
- **SCN5A gene (encodes the alpha subunit of myocardial sodium ion channel)**

Patients on warfarin undergoing emergency surgery: **MUST** give four factor prothrombin complex concentrate.

- If surgery an wait 6 - 8 hours: give 5mg Vitamin K IV

Histology:

- Councilman bodies: Hep C & yellow fever
- Mallory bodies: Alcoholism
- Call-Exner bodies: granulosa cell tumor
- Schiller-Duval bodies: yolk sac tumor

Pulmonary arterial hypertension: exertional dyspnea.

Cranberry juice potentiate anticoagulant effect of Warfarin by inhibiting cytochrome P450 2C9

Second heart sound:

- **Loud: Pulmonary HTN [3x wrong]**
- Soft: AS
- Fixed split: ASD
- **Reversed split (occurs before the first heart sound): LBBB**

Pulses:

- **Alternans:** Seen in HF. upstroke of pulse alternatives between strong and weak, indicating systolic dysfunction.
- Collapsing: forceful rapid upstroke and descent
- Jerky: rapid forceful upstroke
- Slow: slow upstroke
- **Bisferiens: 2 sharp upstrokes during systole**

Third heart sound: heard in:

- constrictive pericarditis
- left ventricular failure (dilated cardiomyopathy) & mitral regurgitation.
- Caused by diastolic filling of the ventricle
- Normal if <30 years or <50 years in women

Temporary pacemaker indications:

- Symptomatic bradycardia not responding to atropine
- Post anterior MI
- Trifascicular block prior to surgery

Mechanism of Valsalva for SVT: after release of strain, BP overshoots above baseline. Baroreflex responds with reflex bradycardia and vasodilation thus returning HR to normal.

Post-ductal variety (upper limb BP > lower limb): aortic coarctation

- associated with neurofibromatosis, bicuspid aortic valve, turner's, and berry aneurysms.

Pulmonary HTN:

Women with pulmonary hypertension should avoid pregnancy due to very high mortality rates.

- Prostacyclins are the primary treatment
- Loud second heart sound
- Diagnosis confirmed with cardiac catheterization

Mixed aortic valve disease is most commonly associated with bisferiens pulse. (Waveform showing 2 systolic peaks and a normal aortic notch) [2x wrong]

Obesity gives false -ve for heart failure.

Recurrent DVT: INR goal of 3.5

INR >8 with no bleeding: Oral Vit K + stop warfarin + repeat in 24hrs

INR >9 with bleeding: Oral Vit K + stop warfarin + repeat in 24hrs

** Warfarin restarted when INR is below 5.

Eisenmenger Syndrome:

CCC: Left to right shunt due to pulmonary hypertension.

Associated with VSD, ASD, PDA

Features: cyanosis, raised JVP, loud second heart sound

Large A Waves in JVP

Aortic Dissection:

- Associated with bicuspid aortic valve, turner, Noonan, pregnant, and Marfan.
- Vague abdominal pain and
- Type A [dissection flap proximal to the brachiocephalic vessels]: ascending aorta (control BP + surgery)

- Type B: descending aorta (control BP)

Carotid artery dissection:

- young, anterior circulation stroke (hemiparesis), head and neck pain, after minor trauma.
- Classic: ipsilateral partial Horner syndrome (ptosis and miosis)
- initial management: anti-thrombotic therapy (aspirin) to prevent thromboembolic complications such as ischemic stroke from thrombus formation at intimal tear site.

Any patient with a drug eluting stent requires a longer duration of clopidogrel treatment.

Gallop rhythm (S3) is a sign of early LVF

New onset A Fib: cardiovert **if 48hrs**

BNP —> suppresses aldosterone secretion. [2X WRONG]

- Secreted by ventricles,
- Acts as a diuretic
- Levels rise in left ventricular failure
- Reduces sympathetic tone
- Acts as a vasodilator

Any major bleeding: stop warfarin, give Vit. K & PCC

V. Tach is precipitated by hypomagnesemia and hypokalemia.

When to stop a TMT?

- BP of 105/70 or less Or 250 systolic or 115 diastolic
- HR 220 minus age

Indications of an implantable cardiac defibrillator:

- Previous MI with non-sustained VT on 24hr monitoring
- HOCM
- Previous cardiac arrest due to V. Fib
- Long QT syndrome

Cardiac syndrome X:

Features of angina (exertional chest pain)

Minimal evidence of atherosclerosis

ECG: downscoping ST depression

Atrial flutter: permanent management is radiofrequency ablation of the tricuspid valve isthmus

- Atrial activity of 300/min + ventricular rate of 150/min

MUGA scan: the most accurate method to determine left ventricular function.

Coronary sinus drains into the right atrium.

Left ventricle receives oxygenated blood from the left atrium (from the pulmonary veins from the lungs)

Right ventricle pumps deoxygenated blood it receives from the right atrium into the pulmonary artery to the lungs.

Inferior vena cava carries deoxygenated blood from the lower half of the body back to the right atrium.

Left atrium receives oxygen rich blood returning from the pulmonary circulation via the 4 pulmonary veins.

Massive pulmonary embolism and hypotension: thrombolyse